

Milestone 5: Type-Indexed Temporal Reputation System

Scoping proposal: chain-health reputation, PRD, and architecture

Scope

Milestone 5 scopes the reputation system with an initial focus on ASI Chain / F1R3FLY chain health. It is a research-synthesis and design milestone, not an implementation milestone. The work synthesizes the current reputation inputs (Meredith's identity-indexed typing judgments, Kolonin's temporal weighted liquid rank, Goertzel's synthesis of the two) together with the tokenomics and shard-health direction from Kevin/Haley into an agreed design direction, documented as a PRD and architecture package that follow-on implementation milestones can be estimated against.

The updated working hypothesis is that the first useful reputation system should collect claims, attestations, provenance, and other evidence about chain actors and infrastructure, then support explicit projections for chain-health consumers. Early personas and reputation-bearing actors should center on validators, while leaving room to extend upward to agents, registered entities, products, services, and artifacts over time as the agent marketplace and broader ecosystem mature.

The milestone should consider the impact across ASI Chain, F1R3FLY, SNET, and users. At a user level, the design should explain what reputation data is collected, how humans or agents provide evidence or claims, how reputation can be queried, and how downstream consumers may use reputation signals without treating any single scalar score as final truth.

The working direction is to hold reputation directly in the F1R3FLY node as a dedicated reputation shard. Evidence is gathered into a reputation graph with full provenance, so invalidating one data point can cascade to everything derived from it. Projections for each chain-health consumer are computed as folds over the graph, with event-driven updates bridged to consuming chains and apps. The milestone should keep the detailed design as a hypothesis to validate, not a fixed implementation commitment.

Deliverables

1. **Research and tokenomics synthesis:** what the reputation papers and Kevin/Haley tokenomics materials imply for chain-health reputation, including shared concepts, divergences, terminology mapping, and recommended design assumptions.

2. **PRD:** what the system must do and for whom - initial personas and consumers, chain-health use cases, the core reputation object model, and requirements for evidence handling, score projection, auditability, disputes, and user/agent interaction.
3. **Evidence and claims model:** a first-pass model for claims, counter-claims, attestations, provenance, evidence levels, and queries such as whether a validator was slashed in a given epoch. This should include how evidence may be collected before final decisions are made about how every signal is used.
4. **Architecture design:** how the system satisfies the PRD - an in-node F1R3FLY reputation-shard design (avoiding sidecar process if possible) covering graph persistence and provenance, projections, event-driven updates and consumer bridges, APIs, identity/scope handling, and the integration boundary with ASI Chain, F1R3FLY, and SNET. The design should mark v1 versus deferred components, including single-shard start versus later multi-shard/federated designs.
5. **Implementation roadmap:** when and in what order - a phased plan starting with evidence collection and transparent projections, then chain-health integrations, hardening, anti-gaming/privacy work, and production readiness.
6. **Open-question register:** a prioritized list of the technical, product, tokenomics, and governance decisions stakeholders must make. These include first personas and consumers, initial reputation types, tokenomics details, the authentication-bundle and key-governance scope, privacy at launch, and how much underlying detail human versus AI consumers can inspect.
7. **Risk register:** prioritized technical, governance, economic, and adversarial risks, including signal quality, gaming, Sybil resistance, validator capture, privacy, disputed attestations, bootstrap trust, and premature coupling to tokenomics policy. Include mitigation notes and identify which risks block implementation versus which can be deferred.

Acceptance Criteria

1. **Research and tokenomics synthesis:** the provided reputation papers and the Kevin/Haley tokenomics materials are consolidated into a single synthesis document that makes shared concepts, divergences, a common terminology mapping, and recommended design assumptions explicit.
2. **PRD:** defines the initial personas and consumers, chain-health use cases, the core reputation object model, and requirements for evidence handling, score projection, auditability, disputes, and user/agent interaction.
3. **Evidence and claims model:** specifies how humans and agents submit evidence (claims, counter-claims, attestations), each with its provenance and evidence level. It also supports queries such as whether a validator was slashed in a given epoch.

4. **Architecture design:** confirms F1R3FLY-node sufficiency and keeps the system in-node (avoiding sidecar processes if possible), and is clear enough for stakeholder sign-off while avoiding over-specification.
5. **Implementation roadmap:** fully fleshes out the next milestone and describes the following two or three at paragraph-level detail, separating near-term work from later hardening, privacy, and production work.
6. **Open-question register:** captures and prioritizes the open technical, product, tokenomics, and governance decisions that require stakeholder input, at enough detail to be actioned.
7. **Risk register:** identifies the major technical, governance, economic, and adversarial risks, plus architecture gaps, with mitigation notes. It also separates the issues that must be resolved before implementation from those that can be monitored, mitigated, or deferred.

References

- L.G. Meredith. Identity-Indexed Typing Judgments and the Adjudication of Capability. Position paper, F1R3FLY Industries, May 2026.
- Anton Kolonin. Generalized Reputation Computation Ontology and Temporal Graph Architecture. arXiv:1912.00176v2; published at MathAI 2025.
- Ben Goertzel. Type-Indexed Temporal Liquid Reputation for ASI:Chain. May 2026.
- Ben Goertzel, Kevin Machiels, Amaury Dallieur, Mario Casiraghi, Nick Nayfack. Next-Gen DeAI Ecosystem: Tokenomics and Shard Economy. December 2025.
- ASI:Chain Reputation - V1/V2 Architecture. Internal sync draft / MLabs Ideas, 22 May 2026.
- Reputation System Vision Overview, rev November 2025.
- Anton Kolonin, Ben Goertzel, Deborah Duong, Matt Ikle. A Reputation System for Artificial Societies. arXiv:1806.07342, 2018. arXiv:1806.07342.